

Highlights

A deterministic multi-agent AI framework for urban heat planning: Coupling district heating simulation, heat-pump grid hosting, and auditable LLM explanations

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- Multi-agent AI couples district heating and heat-pump grid analysis.
- KPI contracts restrict LLM explanations to verified outputs.
- pandapipes and pandapower support street-level assessment.
- Monte Carlo analysis quantifies DH-vs-HP decision robustness.
- The Branitz case demonstrates auditable heat-planning decisions.

A deterministic multi-agent AI framework for urban heat planning: Coupling district heating simulation, heat-pump grid hosting, and auditable LLM explanations

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ARTICLE INFO

Keywords:

Urban heat planning
Multi-agent systems
District heating
Heat pumps
Explainable AI
pandapipes
pandapower

ABSTRACT

Municipal heat planning requires defensible decisions between centralized district heating and decentralized heat-pump deployment, yet conventional workflows often assess heat demand, hydraulic feasibility, low-voltage grid impacts, techno-economics, and stakeholder explanation in separate tools. This paper presents a deterministic multi-agent AI framework for auditable urban heat-planning decisions. The framework couples a Centralized Heating Analysis Agent using pandapipes for district-heating hydraulic and thermal assessment, a Decentralized Heating Analysis Agent using pandapower for heat-pump grid-hosting analysis, and a constrained Urban Heat Decision Coordinator that explains only schema-validated key performance indicators. A deterministic decision cascade first applies pathway-specific feasibility gates. Lifecycle cost, carbon dioxide intensity, and Monte Carlo robustness indicators are then evaluated for technically admissible pathways. The framework is demonstrated for the Branitz district in Cottbus, Germany, using building-level heat demand, street-network geometry, and scenario-based infrastructure assumptions. For the representative ST010 cluster, district heating passed the hydraulic feasibility gate with $\Delta p_{\text{spec}} = 53.0$ Pa/m and $v_{\text{max}} = 0.95$ m/s, whereas the heat-pump pathway failed the low-voltage hosting gate because maximum feeder loading reached 201.5%; the Monte Carlo analysis yielded a district-heating win fraction of $\omega_{\text{DH}} = 1.00$. The contribution is a reproducible energy-AI decision-support workflow that combines physics-based simulation, uncertainty-aware techno-economic assessment, and constrained explainability for municipal heat-planning applications.

1. Introduction

Climate-neutral heat planning requires municipalities to make long-lived infrastructure decisions under technical, economic, and social uncertainty. In Germany, this challenge is framed by national climate targets and municipal heat-planning obligations, while modern heating strategies increasingly require a choice between centralized district heating expansion and decentralized electrification via heat pump systems. This decision is not adequately represented by heat-demand density alone: district heating pathways require hydraulic feasibility, acceptable heat losses, pipe-routing feasibility, and investment justification, whereas heat pump pathways can create new constraints in the low-voltage electricity grid.

Urban heat decarbonization therefore resembles a multi-constraint planning problem rather than a single-objective optimization problem. Prior work on wicked planning problems emphasizes that infrastructure decisions involve conflicting objectives, uncertain boundary conditions, and path dependencies.^{1,2} District heating research has also emphasized the role of fourth-generation district heating systems in future sustainable energy systems.³

Existing planning workflows often remain siloed. district heating feasibility is assessed with thermal-hydraulic models, heat pump deployment is assessed with electrical grid models, and techno-economic comparison is often performed separately. Open-source tools such as pandapipes and pandapower provide strong foundations for reproducible thermal-hydraulic and electrical-grid simulation, respectively.^{4–7} However, the challenge is not only to run these simulators, but to couple their outputs into one auditable decision pipeline.

At the same time, large language models (LLMs) offer new opportunities for stakeholder-oriented explanation, but free-form LLM reasoning is risky in critical infrastructure contexts because generated explanations may contain unsupported claims. Explainable and trustworthy AI methods therefore require architectural constraints that separate deterministic calculation from natural-language synthesis.^{8–10}

This paper proposes a deterministic multi-agent AI framework that couples district heating simulation, heat pump grid-hosting assessment, techno-economic evaluation, uncertainty propagation, and constrained explanation into a single auditable workflow. The framework is demonstrated for the Branitz district in Cottbus, Germany.

The contributions are:

1. a multi-agent architecture that couples pandapipes-based district heating analysis and pandapower-based heat pump grid-hosting assessment on the same street-cluster unit;

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2. a schema-validated KPI contract that separates physical simulation, deterministic decision logic, and LLM-based explanation;
3. a standards-aligned feasibility-gate decision cascade for comparing centralized and decentralized heating pathways;
4. a feasibility-gated lifecycle-cost, CO₂, and robustness assessment that evaluates economic and environmental indicators only for technically admissible pathways; and
5. a reproducible software workflow that links simulation outputs, KPI contracts, decision traces, and open implementation artifacts.

2. Related work

Urban heat planning increasingly requires decisions across coupled technical, economic, environmental, and institutional dimensions. This section reviews five streams of work relevant to the proposed framework: urban heat planning as a multi-constraint decision problem, integrated urban energy models, multi-agent decision support, constrained large language models for critical infrastructure, and verified energy-system simulators for district-heating and low-voltage grid assessment.

2.1. Urban heat planning as a multi-constraint decision

Urban heat planning is no longer a narrow utility-expansion task. It is increasingly understood as a complex socio-technical planning problem involving long asset lifetimes, conflicting stakeholder objectives, uncertain energy prices, regulatory requirements, and deep path dependency. The concept of a “wicked problem”, introduced by Rittel and Webber¹, is particularly relevant because urban heat transitions rarely admit a single correct solution. Each infrastructure choice changes the future planning context by creating physical assets, contractual obligations, and social expectations that are difficult to reverse.

Historically, district-heating expansion was often guided by spatial heat-demand density or linear heat density. These indicators remain useful for first screening, but they are insufficient for final infrastructure decisions. A high heat-density area may still be hydraulically difficult to supply, while a low-density area may become suitable for decentralized electrification if the low-voltage grid has sufficient hosting capacity. Conversely, large-scale heat-pump deployment can appear attractive at building level while producing voltage-band violations or feeder overloads at distribution-grid level. Recent municipal heat-planning guidance therefore emphasizes the need to combine building-stock data, geospatial constraints, existing infrastructure, local energy potentials, and stakeholder considerations¹¹.

The transition toward climate-neutral heating further increases the dimensionality of the decision. Fourth-generation district heating concepts emphasize lower temperatures, reduced losses, renewable integration, and sector coupling³. At the same time, electrification through heat pumps shifts

part of the heating burden to the electricity distribution network. Studies on renewable energy systems and district-level decarbonization show that heating pathways must be assessed not only by delivered heat cost but also by grid impacts, emissions, technology maturity, and long-term flexibility¹². This motivates decision-support methods that move beyond single-metric screening and compare heating options through consistent technical and techno-economic indicators.

2.2. Integrated urban energy models and transparency limitations

Integrated Urban Energy Models (UEMs) have emerged to capture interactions between buildings, geospatial structure, heat networks, electricity networks, and energy demand. Their main advantage is the ability to evaluate cross-sector effects that remain invisible in isolated models. For example, a decentralized heat-pump strategy cannot be judged only by building heat demand; it must also be assessed through low-voltage grid constraints. Similarly, a district-heating strategy cannot be judged only by heat density; it must also satisfy hydraulic pressure, velocity, heat-loss, and routing constraints. Reviews of multi-domain UEMs highlight the importance of combining GIS, building data, network models, meteorological data, and scenario assumptions in a consistent modelling environment¹³.

However, many integrated modelling approaches trade physical fidelity for computational tractability. District-heating networks may be represented through simplified cost-distance approximations, while low-voltage grids may be approximated using aggregate capacity factors rather than explicit power-flow calculations. Such abstractions are useful for early-stage screening but can produce recommendations that fail when engineering constraints are applied. This is particularly problematic in municipal planning, where recommendations must be technically defensible and explainable to non-expert stakeholders.

A second limitation is transparency. As UEMs become more complex, they can become difficult for planners and decision-makers to audit. Black-box optimization, opaque surrogate models, and undocumented assumptions reduce trust in model outputs, especially in public-sector infrastructure decisions. Explainable AI literature has emphasized that transparency is not only a usability feature but also a requirement for accountable decision support in critical domains^{8;14}. One promising response is modular system design, where domain-specific modules produce explicit, inspectable key performance indicators (KPIs) that are later combined through transparent decision rules. This principle motivates the KPI-contract architecture used in the present paper.

2.3. Multi-agent systems in energy decision support

Multi-Agent Systems (MAS) originated from distributed artificial intelligence and cooperative problem solving, where autonomous software entities coordinate to solve tasks that

exceed the capability or knowledge boundary of a single agent^{15;16}. In this sense, MAS provide a natural architecture for decomposing heterogeneous energy-planning tasks into specialized computational agents. Instead of embedding all physical, economic, and explanatory logic into a single monolithic model, MAS assigns responsibilities to domain-specific agents that interact through defined interfaces. In energy systems, MAS and agent-based modelling have been widely applied to smart grids, demand response, microgrid coordination, market simulation, district-level energy management, and multi-energy systems^{17–19}.

Most existing MAS applications in energy systems focus on operational control over short time horizons, such as dispatch, flexibility management, or real-time coordination. Examples include agent-based approaches for district-heating monitoring and control²⁰, power-to-heat coordination²¹, and hierarchical smart-grid management^{22;23}. These applications demonstrate the value of modularity, autonomy, and hierarchical coordination, but they do not fully address strategic municipal heat planning, where the relevant decisions concern long-lived infrastructure investments rather than short-term operation.

Strategic heat planning requires a different MAS design. Agents must compare mutually exclusive heating pathways, apply feasibility constraints, quantify long-term cost and emissions, propagate uncertainty, and generate an auditable recommendation. The time horizon is typically 10–20 years or longer, and the outcome may influence municipal budgets, network investments, and household heating costs. This creates a need for MAS architectures that combine domain simulators, deterministic decision rules, and explainable outputs. The framework proposed in this paper addresses this gap by assigning district-heating simulation, heat-pump grid-hosting analysis, techno-economic evaluation, and explanation generation to separate but coordinated agents.

2.4. Large language models, hallucination risk, and constrained explanation

Large Language Models (LLMs) have created new opportunities for natural-language interfaces, planning assistance, code generation, and explanation of complex technical outputs. In energy systems, recent studies have explored LLMs for grid analytics, documentation, compliance support, and decision assistance^{10;24–26}. Their main advantage is the ability to transform complex structured outputs into accessible explanations for engineers, planners, and non-technical stakeholders.

However, LLMs also introduce a serious risk in critical infrastructure contexts: hallucination. Because LLMs generate plausible text rather than physically verified conclusions, they may produce statements that are numerically incorrect, unsupported by the data, or inconsistent with engineering constraints. In power-system applications, this can include incorrect equipment ratings, invalid technical recommendations, or explanations that contradict simulation outputs¹⁰. For urban heat planning, unsupported AI-generated claims

could mislead decision-makers and weaken the credibility of the planning process.

A safer pattern is therefore to separate calculation, decision-making, and explanation. In the proposed explanation-by-design architecture, the LLM is not responsible for simulating networks, calculating KPIs, modifying assumptions, or selecting the final option. Instead, deterministic simulation and decision agents first generate a schema-validated KPI contract, which records the relevant inputs, assumptions, computed indicators, and decision-relevant outputs. The LLM then operates only as an explanation layer, transforming these already-computed results into a readable narrative for planners and stakeholders. This design follows the broader Explainability-by-Design principle that explanation capability should be embedded into the system architecture through traceable data flows, provenance records, queries, and explanation plans, while also aligning with XAI literature that emphasizes understandable, domain-relevant, and trustworthy explanations for decision support^{8;9}.

For critical-infrastructure applications, grounding alone is not sufficient; generated explanations should also be checked after generation. Hallucination surveys show that neural text generators can produce statements that are fluent but unfaithful to the source evidence²⁷. Retrieval-augmented generation and related grounded-generation methods reduce this risk by conditioning generation on explicit evidence, but they do not eliminate unsupported or contradictory claims²⁸. Therefore, the present framework treats LLM safety as a post-generation verification problem. Generated explanations are decomposed into decision-relevant claims, such as numerical statements, feasibility statements, and comparative statements. These claims are then checked against the structured KPI contract. Supported claims can be retained, unsupported claims trigger rejection or fallback, and contradictory claims are blocked before the explanation is released. This workflow is consistent with claim-verification approaches such as FEVER, which classify claims against evidence as supported, refuted, or not supported, and with hallucination-detection methods such as SelfCheckGPT, which evaluate factual consistency in generated text^{29;30}. Natural Language Inference (NLI) can provide an additional safeguard by classifying generated statements against KPI-derived evidence as entailed, contradicted, or neutral. In this paper, this logic motivates the ClaimExtractor and safety-regression workflow described in Section 3.9.

2.5. Verified simulators for district heating and low-voltage grid analysis

A central requirement for defensible heat-planning recommendations is the use of physically grounded and reproducible simulation tools. In the Python ecosystem, pandapower and pandapipes provide open-source foundations for electrical and pipe-network analysis. pandapower supports power-system modelling, power-flow calculation, and grid analysis using data structures compatible with the wider Python scientific stack⁵. It is particularly suitable for assessing

whether additional heat-pump loads can be hosted by low-voltage networks without violating voltage or equipment-loading constraints.

`pandapipes` provides complementary functionality for fluid and thermal network simulation, including hydraulic calculations, pressure losses, flow velocities, and heat-transfer effects in pipe networks⁴. This makes it suitable for district-heating feasibility assessment, where pipe sizing, pressure-drop limits, thermal losses, and pump-power requirements are central planning indicators. Further developments in both libraries have improved their applicability for integrated multi-energy analysis and cross-domain workflows^{6,7}.

Recent district-heating studies further show the importance of topology, temperature level, and dynamic operation in evaluating future heat networks. Work on district-heating topology co-planning and fourth-generation temperature variation demonstrates that pipe-level modelling can substantially affect feasibility and cost conclusions^{31,32}. Low-temperature district-heating research also shows that network losses and supply-temperature assumptions are central to the economic and environmental performance of district-heating pathways³³. These findings support the use of explicit thermo-hydraulic simulation rather than relying only on aggregate heat-density indicators.

2.6. Standards, hard gates, and research gap

Engineering standards and planning conventions provide objective constraints for determining whether a proposed heating pathway is technically admissible. For district heating, EN 13941-1 defines requirements for the design and installation of buried hot-water district-heating networks, including aspects relevant to pressure, temperature, pipe design, and installation practice³⁴. For low-voltage connections, VDE-AR-N 4100 specifies technical connection requirements relevant to voltage limits, equipment loading, and safe integration of new loads³⁵. Economic appraisal can be supported by AGFW cost conventions, while emissions assessment requires consistent national emission factors such as those reported by the German Environment Agency^{36,37}.

These standards motivate a feasibility-first decision cascade. Candidate options should first pass technical hard gates before being ranked by leveled cost of heat, emissions, or robustness. A heat-pump pathway with low apparent cost but severe low-voltage violations is not a valid recommendation without grid reinforcement. Similarly, a district-heating pathway with excessive pressure drop or unacceptable flow velocities should not proceed to economic ranking as if it were technically feasible.

The reviewed literature provides important components: urban heat-planning methods, integrated energy models, MAS architectures, LLM-based explanation, and verified simulators. However, these components are rarely combined into a single auditable workflow that compares district heating and decentralized heat pumps at street-cluster scale using the same geospatial demand basis. In particular, there

remains a gap for a deterministic framework that jointly evaluates district-heating thermo-hydraulics, heat-pump low-voltage grid hosting, techno-economic performance, emissions, uncertainty, and constrained natural-language explanation. This paper addresses that gap by proposing a multi-agent AI framework that couples `pandapipes`, `pandapower`, deterministic KPI contracts, feasibility-first decision rules, Monte Carlo robustness analysis, and read-only LLM explanation for municipal heat-planning support.

3. Methodology

This section presents the proposed deterministic multi-agent AI framework for street-cluster-scale urban heat planning. The methodology combines physics-based simulation, techno-economic evaluation, uncertainty analysis, and constrained natural-language explanation. The central design principle is that numerical computation and decision logic are separated from the language model: the Large Language Model (LLM) is allowed to explain validated results, but it is not allowed to simulate, optimize, modify assumptions, or decide between heating pathways.

The framework evaluates two mutually exclusive infrastructure strategies for a given street cluster: centralized district heating (DH) and decentralized heat pumps (HPs). The DH pathway is assessed through thermo-hydraulic simulation using `pandapipes`, while the HP pathway is assessed through low-voltage grid-hosting analysis using `pandapower`. The outputs of these domain agents are written into a schema-validated Key Performance Indicator (KPI) contract, which is then processed by a deterministic decision engine and a read-only explanation layer.

3.1. Street-cluster decision formulation

The basic unit of analysis is the street cluster. A street cluster is defined as

$$S = \{B, G, D\}, \quad (1)$$

where $B = \{b_1, \dots, b_n\}$ denotes the set of buildings in the cluster, $G = (V, E)$ denotes the street-network graph derived from geospatial data, and D denotes the district context. The district context includes information such as proximity to potential heat-supply points, street-level routing constraints, low-voltage grid assumptions, and relevant planning parameters.

For each street cluster, the framework compares two heating pathways. The centralized pathway connects the cluster to a district-heating network:

$$A : S \rightarrow \text{DH}, \quad (2)$$

whereas the decentralized pathway assigns an individual heat pump to each building and connects the resulting electrical loads to the low-voltage grid:

$$B : S \rightarrow \{\text{HP}_i\}_{i=1}^n \rightarrow \text{LV grid}. \quad (3)$$

The output of the decision process is defined as

$$R = \{F, K, \delta\}, \quad (4)$$

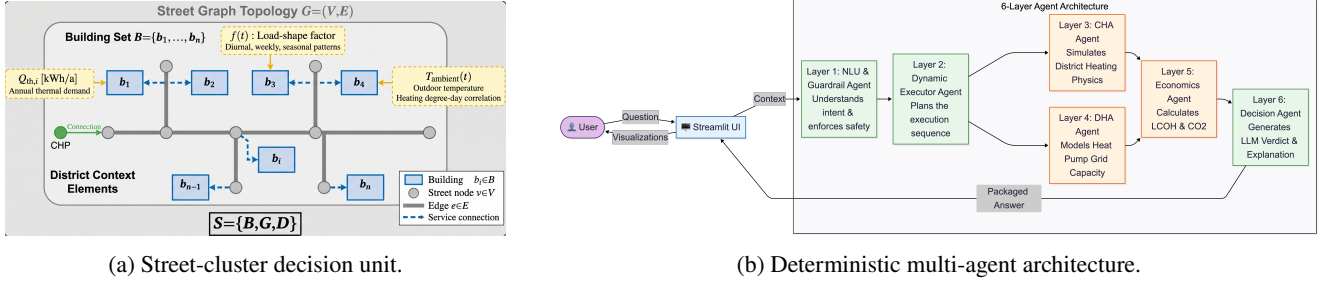


Figure 1: Street-cluster formulation and deterministic multi-agent architecture used in the proposed framework. (a) The street-cluster decision unit defines the common spatial basis for comparing district-heating and heat-pump pathways. (b) The multi-agent architecture separates physical simulation, economic calculation, deterministic decision logic, and constrained LLM-based explanation.

where F contains feasibility flags, K is the KPI vector, and δ is the final recommendation:

$$\delta \in \{\text{DH, HP, Hybrid, Inconclusive}\}. \quad (5)$$

The street-cluster scale is selected because it balances municipal interpretability and technical resolution. A whole-district model can obscure local grid bottlenecks and pipe-routing constraints, while a purely building-level model misses network economies of scale. The street cluster therefore provides a practical decision unit for comparing infrastructure alternatives in a form that is meaningful for both engineering analysis and municipal planning.

3.2. Multi-agent system architecture

The framework is organized as a deterministic Multi-Agent System (MAS). Each agent is responsible for a clearly bounded task and exchanges information through structured outputs rather than unstructured text. This avoids a monolithic architecture and makes each step of the workflow auditable.

The proposed system consists of six functional components. The Data Preparation Agent harmonizes building data, heat-demand data, street geometry, and scenario assumptions. The Centralized Heating Analysis Agent (CHA) evaluates the district-heating option using pandapipes. The Decentralized Heating Analysis Agent (DHA) evaluates the heat-pump option using pandapower. The Economics Agent computes lifecycle cost and emissions indicators. The Decision Agent applies deterministic feasibility and ranking logic. Finally, the Urban Heat Decision Coordinator (UHDC) produces a stakeholder-readable explanation from the KPI contract.

The architecture uses a feasibility-first sequence. Domain agents first determine whether each pathway is technically admissible. Only technically feasible candidates are passed to economic and environmental ranking. This prevents a pathway with attractive cost but violated hydraulic or electrical constraints from being recommended as a valid planning option.

3.3. Data preparation and demand representation

The Data Preparation Agent constructs the common input basis used by all downstream agents. Building-level

heat demand, geospatial building geometry, street-network geometry, weather-related assumptions, and scenario parameters are harmonized into a consistent street-cluster representation. This common data basis is essential because DH and HP pathways must be evaluated for the same buildings, the same heat demand, and the same planning boundary.

Hourly heat demand for building i at time t is represented as

$$Q_{th,i}(t) = Q_{ann,i} \cdot f_i(t) \cdot \gamma(t), \quad (6)$$

where $Q_{ann,i}$ is the annual thermal demand, $f_i(t)$ is the normalized temporal load-profile factor, and $\gamma(t)$ is a weather- or degree-day-related correction factor. The resulting demand profiles are used to identify representative peak and stress hours for both the district-heating and heat-pump analyses.

The key methodological requirement is consistency: the CHA and DHA agents must evaluate infrastructure options under the same thermal demand conditions. This allows pressure-drop, heat-loss, voltage, feeder-loading, LCOH, and emissions indicators to be compared within one coherent KPI contract.

3.4. Centralized Heating Analysis Agent

The Centralized Heating Analysis Agent evaluates the DH pathway. Its purpose is to synthesize a street-following district-heating layout and determine whether it can deliver the required heat while satisfying hydraulic and thermal constraints. The agent uses pandapipes, an open-source pipe-network simulation tool suitable for hydraulic and thermal calculations in district-heating and gas networks^{4,6}.

For a thermal load Q , the required mass flow is approximated as

$$\dot{m} = \frac{Q}{c_p \Delta T \eta}, \quad (7)$$

where $\dot{m}$ is mass flow, c_p is the specific heat capacity of the fluid, ΔT is the design temperature difference between supply and return, and η represents efficiency assumptions. This relation links building-level heat demand to pipe-level hydraulic requirements.

The CHA follows four steps. First, it maps buildings to the street graph and generates a street-following network

Table 1

Agent roles, tools, inputs, and outputs in the proposed framework.

Agent	Tool / method	Main inputs	Main outputs
Data Preparation Agent	GIS and preprocessing scripts	Building data, heat demand, street geometry, scenario assumptions	Cleaned building set, street clusters, demand profiles
Centralized Heating Analysis Agent	pandapipes	Street graph, heat demand, pipe assumptions, supply-temperature assumptions	Velocity, pressure-drop indicators, heat losses, pump power, DH feasibility
Decentralized Heating Analysis Agent	pandapower	HP loads, COP assumptions, base electrical load, LV-grid model	Feeder loading, transformer loading, voltage range, grid headroom, HP feasibility
Economics Agent	LCOH and emissions model	CAPEX, OPEX, energy prices, discount rate, emission factors	LCOH, CO ₂ intensity, lifecycle indicators, uncertainty distributions
Decision Agent	Rule-based cascade	Feasibility flags and KPI contract	DH/HP/Hybrid/Inconclusive recommendation
Urban Heat Decision Coordinator	Constrained LLM	Schema-validated KPI contract	Auditable natural-language explanation

layout. Second, it assigns heat loads to demand nodes and estimates design mass flows. Third, it runs hydraulic and thermal simulation in pandapipes. Fourth, it writes the resulting DH indicators to the KPI contract.

The main CHA outputs are total pipe length, maximum flow velocity, pressure-drop indicators, heat-loss percentage, pump-power requirement, and a binary feasibility flag. The feasibility interpretation is aligned with district-heating design practice and EN 13941-1, which provides requirements for buried district-heating networks³⁴. In this paper, these constraints are used as planning-level hard gates rather than as a replacement for detailed engineering design.

3.5. Decentralized Heating Analysis Agent

The Decentralized Heating Analysis Agent evaluates the HP pathway. Its purpose is to determine whether the low-voltage grid can host a decentralized heat-pump rollout serving the same thermal demand used in the DH analysis. The agent uses pandapower, an open-source power-system analysis tool for load-flow and grid assessment^{5,6}.

For each building i , thermal demand is converted into electric heat-pump load using

$$P_{HP,i}(t) = \frac{Q_{th,i}(t)}{COP_i(t)}, \quad (8)$$

where $P_{HP,i}(t)$ is the electric power demand of the heat pump and $COP_i(t)$ is the coefficient of performance. The total electric demand at time t is then

$$P_{total}(t) = P_{base}(t) + \sum_{i=1}^n P_{HP,i}(t), \quad (9)$$

where $P_{base}(t)$ is the non-heating base electrical demand.

The DHA runs load-flow simulations for selected stress hours and reports voltage profiles, feeder loading, transformer loading, violation counts, and available grid headroom. The planning interpretation is aligned with low-voltage technical connection requirements such as VDE-AR-N 4100³⁵. If the HP pathway causes voltage violations or

excessive loading, the pathway is flagged as infeasible unless reinforcement is explicitly included as a scenario.

3.6. Techno-economic and emissions model

The Economics Agent compares feasible alternatives using the Levelized Cost of Heat (LCOH). The LCOH is defined as

$$LCOH = \frac{\sum_{t=0}^T \frac{C_{inv}(t) + C_{op}(t)}{(1+r)^t}}{\sum_{t=1}^T \frac{E_{delivered}(t)}{(1+r)^t}}, \quad (10)$$

where $C_{inv}(t)$ denotes investment cost, $C_{op}(t)$ denotes operating cost, r is the discount rate, T is the planning horizon, and $E_{delivered}(t)$ is delivered heat. Cost assumptions are structured using district-heating cost conventions and scenario parameters³⁶.

Specific emissions are calculated as

$$I_{CO_2} = \frac{\sum_f E_f \phi_f}{E_{total}}, \quad (11)$$

where E_f is the energy consumption of carrier f , ϕ_f is the corresponding emission factor, and E_{total} is the total delivered heat. National inventory and emission-factor sources are used to maintain consistent baseline assumptions³⁷.

The economic model is not used to override technical infeasibility. Instead, LCOH and emissions are evaluated after the feasibility gates. This ensures that a technically invalid option is not selected because of a favorable cost estimate.

3.7. Uncertainty and robustness analysis

Long-term heat-planning decisions are sensitive to uncertain inputs such as heat demand, electricity price, heat-pump COP, pipe capital cost, discount rate, and network heat losses. The framework therefore includes a Monte Carlo robustness layer. For each sample s , uncertain parameters are drawn from predefined scenario distributions, and the

Table 2

Example KPI contract fields used by the deterministic decision engine and constrained explanation layer.

KPI group	Example fields	Decision role
DH feasibility	Maximum velocity, pressure-drop indicator, heat losses, pump power	Filters infeasible DH solutions before cost comparison
HP grid hosting	Feeder loading, transformer loading, minimum voltage, violation count	Filters infeasible HP rollout scenarios before cost comparison
Economics	LCOH, CAPEX, OPEX, planning horizon, discount rate	Compares technically feasible alternatives on lifecycle cost
Emissions	CO ₂ intensity, energy-carrier assumptions, emission factors	Supports climate comparison and tie-breaking
Robustness	Monte Carlo win fraction, confidence interval, sensitivity drivers	Tests decision stability under uncertainty
Explanation safety	Supported claims, rejected claims, fallback-response flag	Prevents unsupported natural-language statements

decision engine evaluates a corresponding recommendation δ_s .

Decision stability is summarized using the DH win fraction:

$$\omega_{\text{DH}} = \frac{1}{N} \sum_{s=1}^N \mathbb{1}\{\delta_s = \text{DH}\}, \quad (12)$$

where N is the number of samples and $\mathbb{1}\{\cdot\}$ is the indicator function. A value close to 1 indicates that DH is consistently selected under uncertainty, while a value close to 0 indicates that HP is consistently selected. Intermediate values indicate decision sensitivity and require further planning interpretation.

The robustness layer supports two planning functions. First, it identifies whether the final recommendation is stable under plausible parameter changes. Second, it reveals which assumptions drive possible decision reversals, such as electricity price, COP, pipe cost, or demand intensity.

3.8. KPI contract and constrained LLM explanation

All numerical agents write their outputs into a schema-validated KPI contract. The KPI contract is the only information boundary between deterministic computation and natural-language explanation. It can be represented as

$$K = \{K_{\text{phys}}, K_{\text{grid}}, K_{\text{econ}}, K_{\text{CO}_2}, K_{\text{robust}}, K_{\text{decision}}\}, \quad (13)$$

where K_{phys} contains DH hydraulic and thermal indicators, K_{grid} contains HP grid-hosting indicators, K_{econ} contains lifecycle cost indicators, K_{CO_2} contains emissions indicators, K_{robust} contains uncertainty results, and K_{decision} contains the final decision trace.

The detailed LLM explanation and safety-regression workflow is described in Section 3.9.

3.9. LLM-based explanation and safety-regression workflow

The Urban Heat Decision Coordinator (UHDC) provides the natural-language interface between the deterministic planning workflow and the human user. Its role is not

to perform simulation, optimize a heating pathway, retrieve external information, or modify scenario assumptions. Instead, the UHDC receives a user query, selects the relevant schema-validated KPI contract, generates a bounded explanation from the verified fields, and applies a safety-regression workflow before the answer is released.

The interaction begins with a user question such as why a specific pathway is recommended, which constraint failed, or whether an alternative scenario should be evaluated. The UHDC maps this question to the corresponding street cluster, scenario identifier, and KPI groups. For example, a question about the ST010 recommendation retrieves the feasibility, grid-hosting, economic, emissions, robustness, and decision-trace fields associated with that cluster. The selected KPI contract is then passed to the LLM together with a system instruction that restricts the answer to the supplied fields. The LLM is explicitly prohibited from inventing numerical values, changing feasibility flags, adding unmodelled assumptions, or presenting an excluded pathway as technically admissible.

The bounded explanation process is therefore evidence-first. The deterministic agents produce the evidence, the Decision Agent produces the recommendation, and the LLM only translates the result into stakeholder-readable language. The generated response must refer only to supported fields such as hydraulic feasibility, pressure-drop indicators, maximum velocity, low-voltage loading, voltage range, levelized cost of heat, CO₂ intensity, Monte Carlo win fraction, and the final decision trace. This follows an explanation-by-design approach in which the explanation layer is architecturally constrained by traceable data flows and verified intermediate artifacts^{9,10}.

To reduce unsupported claims, the framework applies an internal safety-regression workflow after generation. The first stage is a deterministic ClaimExtractor. It scans the generated answer and identifies decision-relevant claims, including numerical claims, comparative claims, and feasibility claims. Examples include statements such as “district heating passed the hydraulic gate”, “maximum feeder loading reached 201.5%”, or “heat pumps are impossible”. Each

Table 3

Example constrained LLM interactions for ST010. The answer is generated only from fields available in the schema-validated KPI contract.

User question	Retrieved KPI fields	Constrained answer behaviour	Safety check
Why is district heating recommended for ST010?	DH feasibility = PASS; HP feasibility = FAIL; maximum feeder loading = 201.5%; DH LCOH = 64.64 EUR/MWh; $\omega_{DH} = 1.00$	The answer explains that district heating is recommended because it passes the hydraulic feasibility gate, while the baseline heat-pump rollout fails the low-voltage grid-hosting gate due to feeder overloading.	Supported
Is the heat-pump option impossible?	HP feasibility = FAIL; failure reason = feeder overloading; reinforcement scenario = not modelled	The answer avoids saying that heat pumps are impossible. It states that the baseline full-HP rollout is not admissible without grid reinforcement and that a reinforced-grid HP scenario should be evaluated separately.	Corrected
Which constraint blocks the heat-pump pathway?	Minimum voltage = 0.9658 p.u.; maximum voltage = 1.0209 p.u.; maximum feeder loading = 201.5%; overloaded lines = 12	The answer identifies feeder loading as the binding constraint and clarifies that voltage and transformer checks do not cause the failure in the baseline scenario.	Supported
Can the LLM change the recommendation?	Decision source = deterministic cascade; final recommendation = DH; LLM access mode = read-only	The answer states that the LLM cannot change the recommendation because the decision is produced by deterministic rules over the KPI contract.	Supported

extracted claim is normalized into a structured representation containing the claim type, target pathway, metric name, numerical value if present, and polarity.

The second stage checks the extracted claims against the KPI contract. Numerical claims are compared against the corresponding KPI field using a tolerance rule. Feasibility claims are checked against the stored feasibility flags. Comparative claims are checked against the deterministic decision trace. If a generated answer states that the HP pathway is feasible while the KPI contract records an HP grid-hosting failure, the answer is rejected or replaced by a fallback response. Similarly, if the LLM claims that heat pumps are technically impossible, the answer is corrected because the KPI contract only supports the narrower statement that the baseline full-HP rollout is not admissible without grid reinforcement.

A further contradiction-checking stage can be implemented using Natural Language Inference (NLI). In this stage, the generated explanation is compared against evidence statements derived from the KPI contract and classified as entailed, contradicted, or unsupported. In the present implementation, deterministic claim extraction and KPI-field checking form the core safety-regression test, while NLI-based contradiction checking is treated as an extensible second-stage safeguard for future deployments. The fallback policy is conservative: if an answer contains an unsupported or contradictory claim, the system either removes the claim, regenerates the answer using stricter instructions, or returns a short statement that the requested conclusion is not supported by the current KPI contract.

Table 3 illustrates the intended interaction pattern. The answers shown are not independent LLM decisions; they

are natural-language renderings of already-computed KPI evidence.

3.10. Deterministic decision cascade

The Decision Agent applies a deterministic feasibility-first cascade. The candidate pathways are first filtered by technical feasibility:

$$\mathcal{A}_{\text{feas}} = \{a \in \{\text{DH}, \text{HP}\} : F_a = 1\}. \quad (14)$$

If no pathway is feasible, the recommendation is marked as inconclusive or requires infrastructure reinforcement. If only one pathway is feasible, that pathway is recommended. If both pathways are feasible, the decision proceeds to LCOH comparison:

$$\delta = \arg \min_{a \in \mathcal{A}_{\text{feas}}} \text{LCOH}_a. \quad (15)$$

If the cost difference falls within a predefined indifference band, the decision uses secondary criteria such as CO₂ intensity, robustness, and reinforcement requirements. This cascade makes the decision trace auditable because every recommendation can be reconstructed from ordered rules and verified KPI fields.

This design prevents the LLM from becoming a hidden decision-maker. The final recommendation is produced by deterministic rules over validated physical, economic, and uncertainty indicators. The LLM only communicates the result in natural language.

3.11. Reproducibility and implementation boundary

The implementation is designed for reproducibility and auditability. Each agent receives explicit input parameters

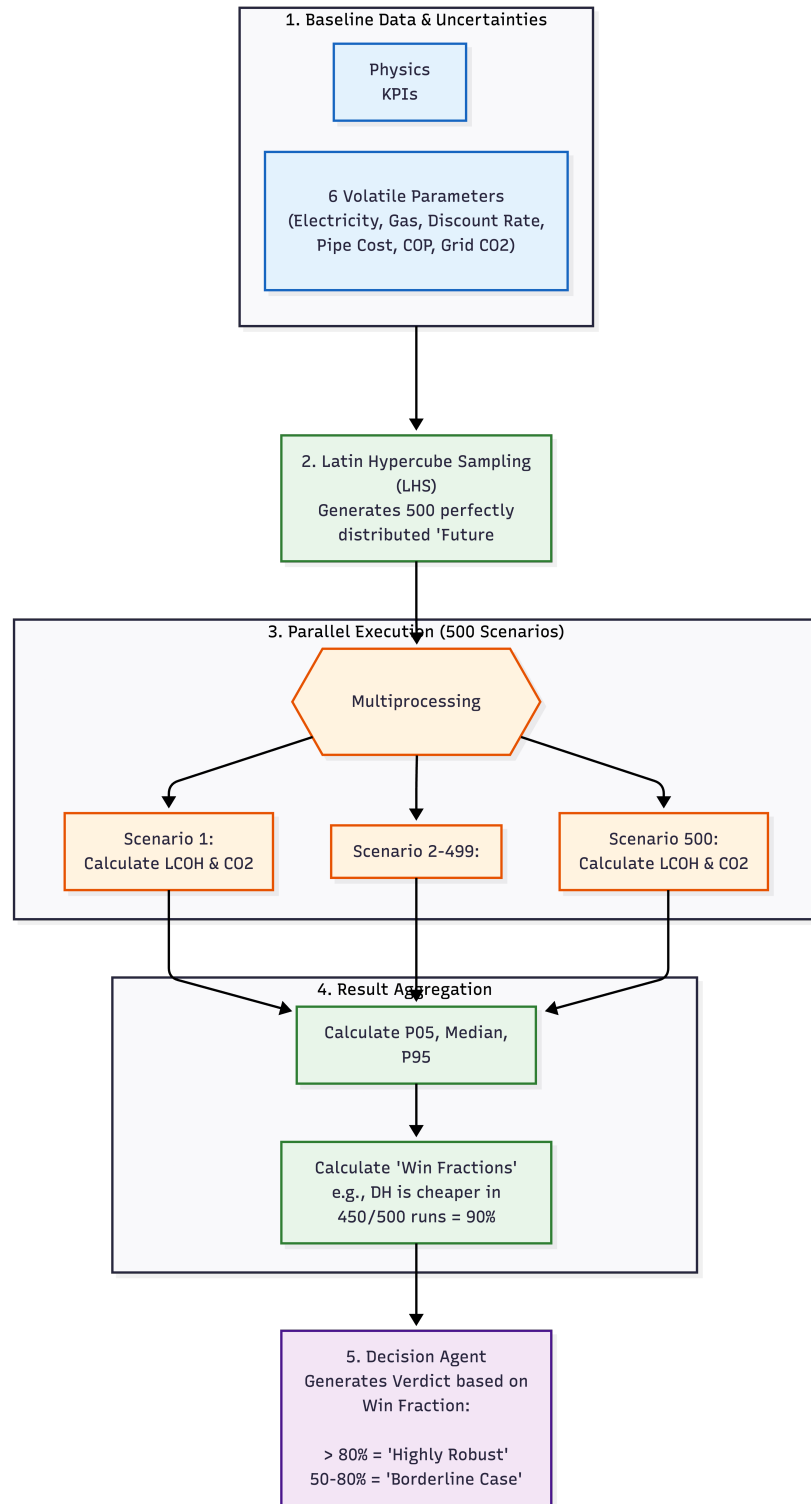


Figure 2: Uncertainty and robustness workflow. Each Monte Carlo sample modifies uncertain parameters, re-evaluates the feasible heating pathways, and records the resulting recommendation. Decision stability is summarized using the win fraction and supporting distributional indicators.

and writes structured outputs. Scenario assumptions, solver versions, and key parameters are stored with the resulting KPI contract. Where stochastic sampling is used, fixed seeds or reproducible sampling configurations are applied. This allows repeated executions of the same scenario to produce the same decision trace, which is essential for public-sector planning and scientific verification. The framework boundary is intentionally conservative. It is not a replacement for detailed engineering design, utility-grade grid studies, or final municipal procurement decisions. Instead, it is a reproducible decision-support layer that helps planners compare DH and HP pathways consistently at street-cluster scale before advancing selected options to detailed engineering assessment.

4. Case study and data

The proposed framework is demonstrated for the Branitz district in Cottbus, Brandenburg, Germany. Branitz is used as a representative municipal heat-planning case because it contains residential street clusters with heterogeneous building geometry, local heat-demand variation, and infrastructure choices that are relevant to the broader district-heating versus heat-pump planning problem. The case study evaluates heating pathways at street-cluster scale rather than only at whole-district or individual-building scale. This spatial resolution is sufficiently fine to capture local pipe-routing and low-voltage grid constraints, while remaining compatible with municipal planning and communication workflows.

The empirical case-study workflow combines four data categories: building-level heat-demand information, OpenStreetMap-derived street and building geometry, planning-grade low-voltage grid representations, and scenario assumptions for district-heating supply and decentralized heat-pump deployment. All data are transformed into a common street-cluster representation before being passed to the simulation agents. This ensures that the Centralized Heating Analysis Agent and the Decentralized Heating Analysis Agent evaluate the same buildings, the same heat demand, and the same spatial boundary.

4.1. Study area

The analysed study area covers the Branitz district of Cottbus with a convex-hull area of approximately 1.46 km². The raw geospatial basis contains 2,079 OpenStreetMap building footprints. After filtering for residential buildings with valid heat-demand information, 550 buildings remain in the demand dataset. These buildings form the planning population for street-cluster generation and downstream pathway comparison.

The street-cluster generation procedure groups buildings by normalized street-name affinity and spatial proximity. Buildings are assigned to the same cluster when they share the same normalized street-name token and lie within a 50 m Euclidean distance threshold. This procedure generates 24 street clusters, denoted ST001–ST024, covering 540 of the 550 valid-demand residential buildings. The remaining 10 buildings are isolated beyond the clustering threshold and

are excluded from network synthesis to avoid unrealistic service-connection assumptions.

The detailed evidence chain in this paper focuses on ST010, Heinrich-Zille-Straße, because it provides a compact but technically meaningful street-level example. In the processed thesis dataset, ST010 contains 73 mapped residential buildings. This cluster is used to demonstrate how the same demand basis is passed through the district-heating, heat-pump grid-hosting, techno-economic, uncertainty, and explanation modules.

4.2. Input data

Table 4 summarizes the main input data categories and modelling assumptions used in the Branitz case study. The Branitz geospatial and energy-data preparation builds on prior preprocessing work for a digital-twin learning environment by Güler³⁸. In that work, OpenStreetMap extraction, low-voltage network generation, LOD2 and ALKIS building-data integration, heat-demand calculation, and pandapower/pandapipes simulation workflows were developed for the Branitzer Siedlung. The present paper reuses this prepared Branitz data basis as a planning-grade input layer and extends it with the deterministic multi-agent decision workflow, KPI-contract structure, constrained LLM explanation layer, and robustness analysis. Street and building geometries originate from OpenStreetMap and are used for cluster generation, map matching, and district-heating network synthesis. Because utility-grade low-voltage grid asset data are confidential and not available for this research, the heat-pump hosting analysis uses planning-grade synthetic radial feeder models. These models are not intended to replace utility-calibrated grid studies; instead, they provide a reproducible basis for comparing decentralized heat-pump scenarios against voltage and loading constraints.

4.3. Preprocessing and street-cluster generation

The Data Preparation Agent transforms raw geospatial and heat-demand records into a consistent simulation environment. This step is necessary because raw municipal and OpenStreetMap records are not directly suitable for physical network simulation. Building polygons, street centre-lines, heat-demand attributes, and cluster identifiers must be aligned before they can be used in pandapipes or pandapower.

The preprocessing pipeline consists of four steps. First, building footprints are filtered to retain residential buildings with valid demand data. Second, buildings are grouped into street clusters using normalized street-name matching and a 50 m proximity threshold. Third, each building centroid is orthogonally projected onto the nearest street segment to define a reproducible service-connection point. Fourth, the processed data are materialized as stable artifacts for downstream agents.

The clustering rule is expressed as

$$b_i \sim b_j \iff \text{name}(b_i) = \text{name}(b_j) \wedge \text{dist}(b_i, b_j) \leq 50 \text{ m}, \quad (16)$$

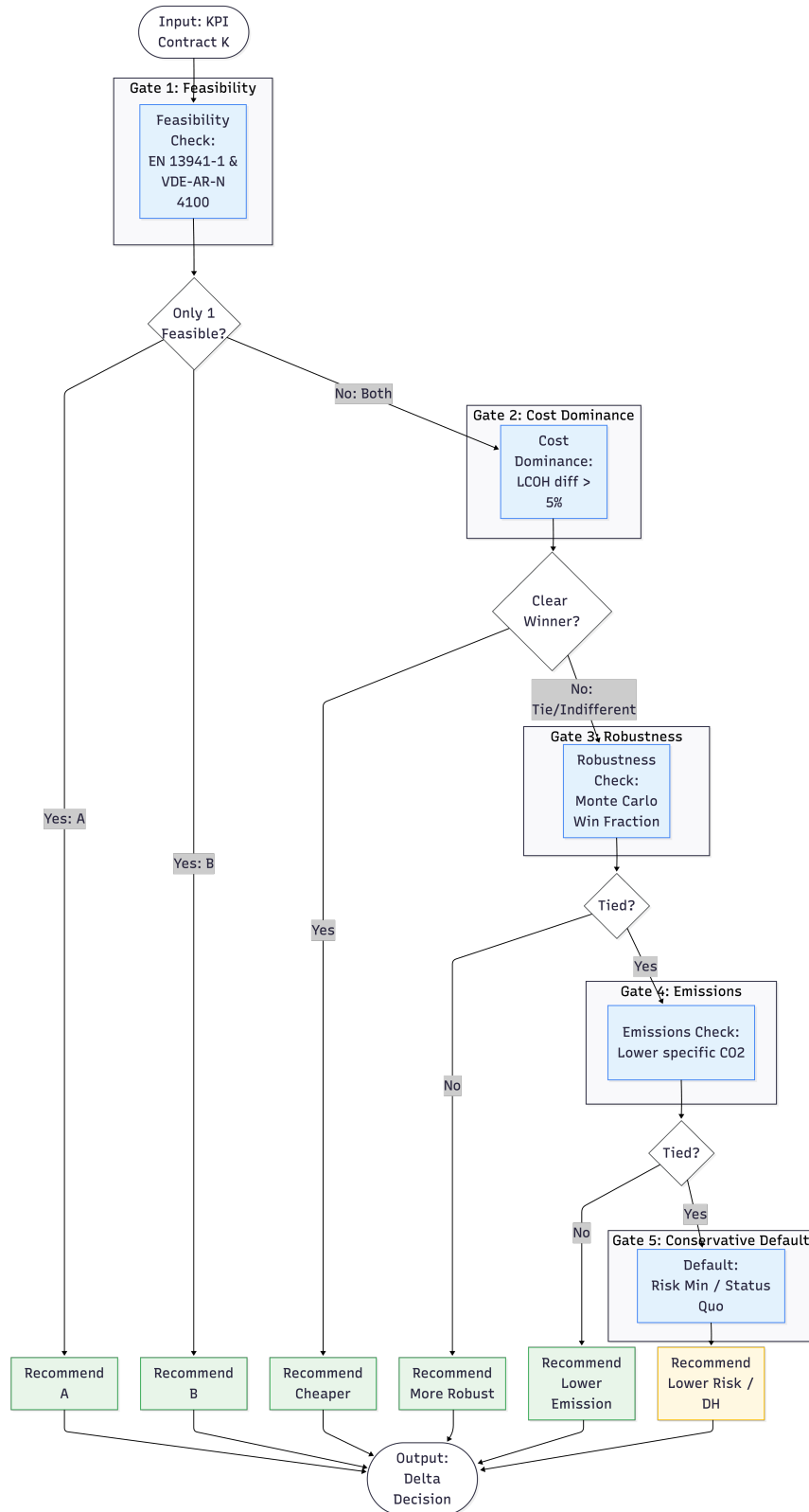
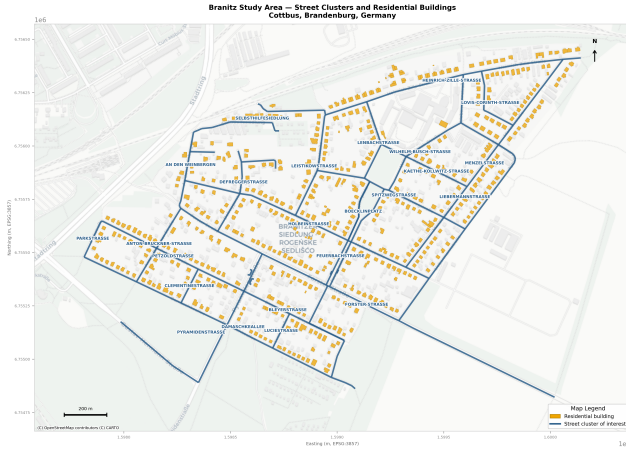
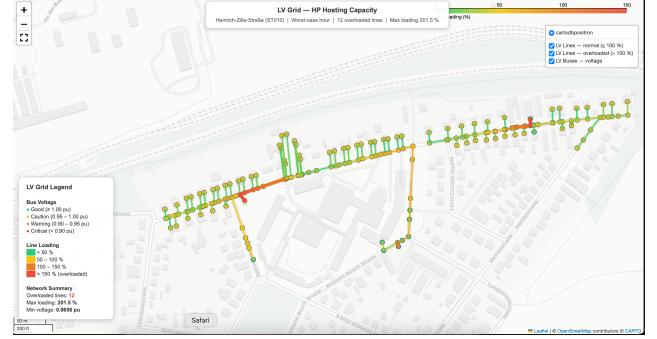


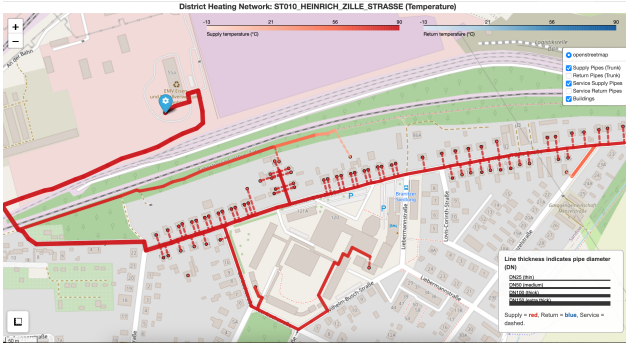
Figure 3: KPI-contract-driven decision cascade. Technical feasibility gates are evaluated before lifecycle cost, emissions, and robustness criteria. The LLM explanation layer receives only the final KPI contract and decision trace.



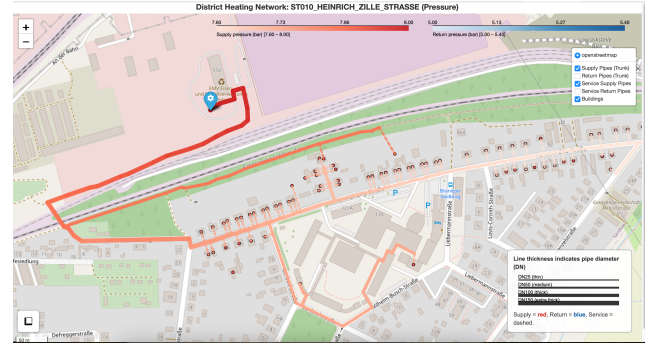
(a) Case-study area of the Branitzer Siedlung in Cottbus, Brandenburg.



(b) Low-voltage grid hosting-capacity result for the heat-pump scenario.



(c) District-heating network temperature distribution.



(d) District-heating network pressure distribution.

Figure 4: Branitz case-study area and comparative infrastructure visualisation for ST010. (a) Residential buildings and street-cluster structure used for the Branitz preprocessing and clustering workflow. (b) Heat-pump pathway evaluated through the low-voltage grid hosting-capacity map. (c) District-heating network temperature distribution. (d) District-heating pressure distribution. Together, the four panels show how the framework links the spatial case-study basis with electrical-grid and thermo-hydraulic infrastructure assessment at street-cluster scale. Basemap and street-network data: OpenStreetMap contributors; visualization prepared by the authors.

where b_i and b_j are buildings, $\text{name}(\cdot)$ denotes the normalized street-name token, and $\text{dist}(\cdot)$ denotes Euclidean distance. A separate service-connection snap threshold of 30 m is used to avoid unrealistic links between buildings and distant network segments. Buildings exceeding this snap threshold are classified as unconnectable under the planning-grade assumptions.

The resulting processed artifacts include building-level geometry and demand tables, street-cluster metadata, hourly heat profiles, and scenario configuration files. These artifacts are treated as write-once inputs to the simulation agents. This design supports reproducibility because any change to raw data, preprocessing parameters, or scenario assumptions produces a distinct processed dataset and therefore a traceable simulation state.

4.4. Hourly heat-demand profiles and stress-hour selection

Infrastructure planning must verify performance under stress conditions rather than average load. District-heating networks must satisfy hydraulic and thermal constraints during high thermal demand, while heat-pump rollout scenarios must satisfy low-voltage grid constraints during coincident electrical peaks. Therefore, the case study uses 8760-hour heat-demand profiles and a Top-N stress-hour selection method.

For each building i , the annual heat demand is transformed into an hourly demand trajectory:

$$q_i(t) = Q_{\text{th},i} \cdot f_{\text{shape}}(t) \cdot g_{\text{temp}}(T_{\text{amb}}(t)), \quad t \in \{1, \dots, 8760\}, \quad (17)$$

where $Q_{\text{th},i}$ is the annual thermal demand of building i , $f_{\text{shape}}(t)$ is the diurnal and weekly load-shape factor, and $g_{\text{temp}}(T_{\text{amb}}(t))$ is the temperature-dependent heating correction.

Table 4

Case-study data sources and modelling assumptions for the Branitz analysis.

Category	Source / representation	Coverage	Workflow role and limitation
Building heat demand	Branitz preprocessing dataset based on municipal Wärmekataster-level records and prior data-integration work by Güler ³⁸	550 residential buildings with valid heat-demand data after filtering 2,079 raw building footprints	Provides annual thermal demand and basic typology; end-use granularity is limited by the available planning dataset
Street and building geometry	OpenStreetMap-derived GeoJSON layers, LOD2 building models, and ALKIS-derived building/address data prepared in the Branitz preprocessing workflow ³⁸	Branitz study area, approximately 1.46 km ²	Used for clustering, map matching, and district-heating routing; complex intersections may require manual correction
Street clusters	Deterministic preprocessing from building and street data	24 clusters, ST001–ST024; 540 clustered buildings and 10 isolated buildings excluded	Provides the spatial decision unit for DH and HP comparison
Representative cluster	ST010, Heinrich-Zille-Straße	73 mapped residential buildings	Used as the detailed evidence chain for agent execution, feasibility assessment, and decision tracing
Low-voltage grid	Planning-grade synthetic radial feeder model	Cluster-level hosting-capacity assessment	Enables reproducible HP grid-hosting analysis; future work requires utility-calibrated LV data
District-heating supply	Planning-scenario parameters for a Cottbus CHP reference source	ST010 centroid distance approximately 369 m; modelled entry-point distance approximately 481 m	Supports marginal-cost allocation and DH expansion assessment; source characteristics are scenario inputs
Weather and demand profiles	8760-hour weather-linked demand representation	Hourly heat profiles for each processed building	Enables Top-N stress-hour selection for hydraulic and electrical feasibility checks
Uncertainty	Monte Carlo scenario sampling	$N = 500$ samples in the journal evaluation workflow	Tests robustness against demand, COP, energy-price, pipe-cost, loss, and discount-rate uncertainty

For a street cluster S , coincident cluster demand is calculated as

$$Q_{\text{cluster}}(t) = \sum_{i \in S} q_i(t). \quad (18)$$

The hours are then ranked by $Q_{\text{cluster}}(t)$, and the ten highest-demand hours are selected as the stress-hour set $\mathcal{T}_{10}$. This Top-N procedure avoids relying on a single isolated peak and instead evaluates the load plateau that governs network sizing and grid-hosting constraints.

For ST010, the selected design hour is hour 6667, corresponding to 5 October 2023 at 19:00. The coincident cluster heat demand at this hour is 2700.5 kW, and the ambient temperature is -0.7°C . Across the Top-10 stress-hour set, the demand varies only moderately, indicating a stable high-demand plateau rather than a single outlier. Both the district-heating and heat-pump analyses use this common stress-hour basis to ensure comparability between pathways.

4.5. Scenario definitions

The case study uses deterministic base-case assumptions together with uncertainty and structural scenario testing. The deterministic base case defines the reference comparison between the DH and HP pathways. For the DH pathway, the scenario includes the heat-supply reference point, street-following routing assumptions, pipe-sizing constraints, thermal-loss assumptions, and pump-power calculation. For the HP pathway, the scenario includes heat-pump COP assumptions, base electrical load, simultaneity assumptions, and low-voltage hosting constraints.

The uncertainty analysis varies parameters that are expected to influence long-term pathway selection. These include heat demand, electricity price, heat-pump COP, pipe capital cost, discount rate, and district-heating heat losses. Each Monte Carlo sample generates a new parameter realization and re-evaluates the decision cascade. The resulting decision distribution is later used to calculate robustness indicators such as the DH win fraction.

The key methodological point is that scenarios are evaluated through the same deterministic agent workflow. This means that changes in recommendation can be traced to explicit changes in input assumptions rather than to stochastic

Table 5

Top-10 stress hours for ST010, Heinrich-Zille-Straße. The highest-demand hour is used as the primary design hour for hydraulic sizing and heat-pump hosting assessment.

Rank	Hour index	Date and time	T_{amb} (°C)	Cluster demand (kW)
1	6667	2023-10-05 19:00	−0.7	2700.5
2	6434	2023-09-26 02:00	−0.5	2670.8
3	6556	2023-10-01 04:00	−0.2	2634.6
4	7158	2023-10-26 06:00	+0.1	2601.7
5	5735	2023-08-27 23:00	+0.1	2600.8
6	6574	2023-10-01 22:00	+0.1	2592.2
7	5923	2023-09-04 19:00	+0.3	2574.3
8	6481	2023-09-28 01:00	+0.3	2568.4
9	6308	2023-09-20 20:00	+0.4	2567.0
10	6761	2023-10-09 17:00	+0.4	2566.4

Table 6

Scenario structure used for the Branitz case-study evaluation.

Scenario group	Parameters varied	Purpose
Deterministic base case	Reference heat demand, reference COP, reference energy prices, reference pipe cost, reference discount rate	Establishes the main DH-versus-HP comparison under fixed assumptions
Technical stress hours	Top-10 cluster demand hours, low ambient temperature, peak coincident demand	Tests hydraulic and electrical feasibility under design-relevant loading conditions
Monte Carlo uncertainty	Heat demand, electricity price, COP, pipe CAPEX, heat losses, discount rate	Quantifies decision stability and identifies drivers of recommendation changes
Structural what-if cases	Building participation, routing assumptions, reinforcement assumptions, localized grid constraints	Tests counterfactual planning conditions and pathway sensitivity

behaviour of the language model or hidden changes in the decision logic.

4.6. Validation boundary and limitations

The case study is designed as a reproducible planning-grade assessment rather than a final engineering design. Two limitations are especially important. First, exact utility-owned low-voltage network data are not available; therefore, the HP pathway is evaluated using synthetic planning-grade feeder models. These models are suitable for demonstrating the decision workflow and identifying potential hosting-capacity constraints, but final grid-reinforcement decisions require utility-calibrated asset data. Second, real-time operational telemetry is not included. The analysis therefore focuses on steady-state design-hour and Top-N stress-hour feasibility rather than transient operational control.

These limitations are consistent with the intended role of the framework. The system does not replace detailed district-heating design, grid-connection studies, or municipal procurement processes. Instead, it provides an auditable and reproducible screening layer that compares centralized and decentralized heating pathways using a common demand basis, explicit feasibility gates, lifecycle cost indicators, emissions indicators, and uncertainty analysis.

5. Results

This section reports the execution results for the representative Branitz street cluster ST010, Heinrich-Zille-Straße. The results are organized according to the same evidence chain used by the proposed framework: system execution and reproducibility, district-heating feasibility, heat-pump grid-hosting feasibility, techno-economic and emissions comparison, Monte Carlo robustness, deterministic gate tracing, and explanation-safety validation.

The ST010 results are used as a detailed evidence chain because they demonstrate the full multi-agent workflow on one street cluster: the Data Preparation Agent resolves the cluster and demand basis, the Centralized Heating Analysis Agent evaluates the district-heating pathway, the Decentralized Heating Analysis Agent evaluates the heat-pump pathway, the Economics Agent computes lifecycle indicators, and the Decision Agent produces the final recommendation from the KPI contract.

5.1. System execution and reproducibility

The ST010 pipeline executed successfully and generated schema-consistent KPI artifacts for the domain agents and the decision layer. The execution followed the deterministic route defined in the methodology: query interpretation, street-cluster resolution, capability checking, domain-agent

Table 7

Determinism and cache-performance results for the ST010 pipeline.

Component	Fresh runtime	Cached runtime	Speedup	Hash stability
CHA hydraulic simulation	155.4 ± 2.1 s	0.002 ± 0.001 s	77,700×	10/10 identical
DHA power-flow analysis	20.6 ± 0.8 s	0.002 ± 0.001 s	10,300×	10/10 identical
Economics calculation	2.1 ± 0.1 s	0.003 ± 0.001 s	700×	10/10 identical
Total pipeline	178.1 ± 3.0 s	0.007 ± 0.002 s	$\approx 25,000\times$	Bit-identical KPIs

Table 8

District-heating feasibility indicators for ST010.

Metric	Value	Limit	Status
Specific pressure drop	53.0 Pa/m	50–200 Pa/m	PASS
Maximum velocity	0.95 m/s	≤ 1.5 m/s	PASS
Hydraulic feasibility flag	1	1	PASS

execution, KPI-contract construction, and deterministic decision evaluation. Repeated validation runs confirmed that the numerical outputs are reproducible under fixed configuration and fixed Monte Carlo seed.

To verify determinism, the same ST010 decision query was executed ten times. Each run serialized the resulting KPI contract using stable key ordering and computed a SHA-256 hash. All ten runs produced the same KPI-contract hash, confirming bit-identical numerical outputs across repeated executions. Timing varied slightly because of operating-system scheduling and disk I/O, but KPI values, decision codes, feasibility flags, and Monte Carlo samples remained unchanged.

The cache-first workflow reduced repeated ST010 evaluation from approximately three minutes to less than 0.01 s while preserving bit-identical KPI outputs. This is important for municipal planning because scenario workshops often require repeated queries over the same street cluster, alternative KPIs, or modified assumptions. The result shows that the framework can support interactive exploration without sacrificing reproducibility.

5.2. District-heating feasibility

The Centralized Heating Analysis Agent evaluated the district-heating pathway for ST010 using the common design-hour demand basis defined in Section 4.4. The street-following network layout was hydraulically feasible under the adopted planning assumptions. The maximum specific pressure drop was 53.0 Pa/m, which lies inside the adopted planning range of 50–200 Pa/m. The maximum flow velocity was 0.95 m/s, remaining below the 1.5 m/s velocity threshold. The resulting hydraulic feasibility flag was therefore positive:

$$C_{\text{hyd}} = \mathbb{1}\{53.0 \in [50, 200]\} \cdot \mathbb{1}\{0.95 \leq 1.5\} = 1. \quad (19)$$

The DH pathway therefore passes the first technical gate and proceeds to economic and emissions assessment. This

result indicates that, under the baseline network synthesis and pipe-sizing assumptions, a district-heating solution can satisfy the planning-level hydraulic constraints for the ST010 street cluster.

5.3. Heat-pump grid-hosting feasibility

The Decentralized Heating Analysis Agent evaluated whether the same ST010 thermal demand could be supplied by decentralized heat pumps connected to the low-voltage grid. The power-flow results show that the voltage profile remains within the adopted planning band: the minimum bus voltage is 0.9658 p.u. and the maximum bus voltage is 1.0209 p.u. However, the line-loading results reveal a severe feeder-capacity constraint. The maximum loading ratio reaches 201.5%, and 12 distinct lines experience overloads across the ten stress hours.

The heat-pump pathway therefore fails the first technical feasibility gate because of low-voltage feeder overloading. The failure is not caused by voltage-band violation or transformer-capacity violation; instead, the limiting constraint is line loading in the low-voltage feeder network. Under the baseline assumptions, full decentralized heat-pump deployment would require explicit grid reinforcement before it could be treated as a technically feasible option.

5.4. Feasibility-gated cost and emissions assessment

The deterministic decision cascade evaluates lifecycle cost and emissions only after technical feasibility has been checked. For ST010, the baseline full-HP rollout fails the low-voltage grid-hosting gate and is therefore not treated as a technically admissible no-reinforcement option in the economic ranking. The reported LCOH and CO₂ indicators in this section therefore describe the admissible DH pathway under the baseline assumptions, rather than a complete reinforced-grid DH-versus-HP cost comparison.

A full HP lifecycle-cost comparison would require an additional reinforced-grid scenario in which feeder upgrades, implementation constraints, and reinforcement costs are explicitly modelled. This extension is outside the scope of the present baseline case study and is treated as future work. The DH pathway, by contrast, passes the hydraulic feasibility gate and is therefore carried forward to lifecycle-cost and emissions assessment.

The DH pathway achieves an LCOH of 64.64 EUR/MWh and a specific CO₂ intensity of 141.18 kgCO₂/MWh under the adopted baseline assumptions. These values characterize the technically admissible pathway selected by the

Table 9

Low-voltage grid-hosting indicators for ST010 under decentralized heat-pump rollout.

Criterion	Metric value	Planning limit	Status
Minimum bus voltage	0.9658 p.u.	≥ 0.95 p.u.	PASS
Maximum bus voltage	1.0209 p.u.	≤ 1.05 p.u.	PASS
Maximum feeder loading	201.5%	$\leq 80\%$	FAIL
Distinct overloaded lines	12 lines	0	FAIL
Line-overload events	116 events over 10 h	0	FAIL
Transformer violations	0	0	PASS

Table 10

Feasibility-gated cost and emissions assessment for ST010. Economic and emissions indicators are reported only for pathways that pass the technical feasibility gates in the baseline scenario.

Pathway	Baseline feasibility status	LCOH	CO ₂ intensity	Interpretation
District heating	PASS: hydraulic feasibility gate satisfied	64.64 EUR/MWh	141.18 kgCO ₂ /MWh	Technically admissible and carried forward to cost and emissions assessment
Full heat-pump rollout	FAIL: low-voltage grid-hosting gate violated; maximum feeder loading reaches 201.5%	Not ranked	Not ranked	Excluded from the baseline no-reinforcement economic ranking; a reinforced-grid HP scenario is required for full lifecycle-cost comparison

feasibility-first cascade. They should not be interpreted as evidence that all possible HP-based solutions are economically or environmentally inferior. Rather, the result shows that the modelled no-reinforcement HP rollout does not reach the economic ranking stage because it violates the adopted grid-hosting criteria.

The final ST010 recommendation is therefore driven primarily by technical admissibility. Since DH passes the hydraulic gate and the baseline full-HP rollout fails the low-voltage feeder-loading gate, the Decision Agent selects DH as the only admissible baseline pathway. The cost and emissions indicators then provide supporting characterization of the selected DH pathway, while the reinforced-grid HP alternative remains a future scenario requiring additional modelling.

5.5. Monte Carlo robustness

The uncertainty analysis evaluated the robustness of the ST010 recommendation using $N = 500$ Monte Carlo samples with a fixed random seed. The sampled parameters included energy prices, heat-pump COP, pipe-cost multipliers, heat-demand variation, heat-loss assumptions, and the discount rate. For each sample, the framework re-evaluated the decision cascade and recorded the resulting recommendation.

The resulting district-heating win fraction was

$$\omega_{\text{DH}} = 1.00, \quad (20)$$

Table 11

Monte Carlo robustness indicators for ST010.

Indicator	Value
Number of samples	500
Sampling seed	42
DH win fraction, ω_{DH}	1.00
HP win fraction, ω_{HP}	0.00
Recommendation stability	Stable DH recommendation
Primary interpretation	HP blocked by grid feasibility; DH remains admissible

indicating that all sampled futures selected district heating under the implemented decision logic. The complementary heat-pump win fraction was therefore

$$\omega_{\text{HP}} = 0.00. \quad (21)$$

The robustness result confirms that the ST010 recommendation is not a fragile point-estimate outcome. The decision is stable because the HP pathway is already constrained by the technical feasibility gate, while DH remains admissible under the baseline hydraulic assessment. In planning terms, the result indicates that the main uncertainty for this cluster is not whether an unconstrained HP option is marginally cheaper, but whether grid reinforcement would be introduced as a separate scenario and costed explicitly.

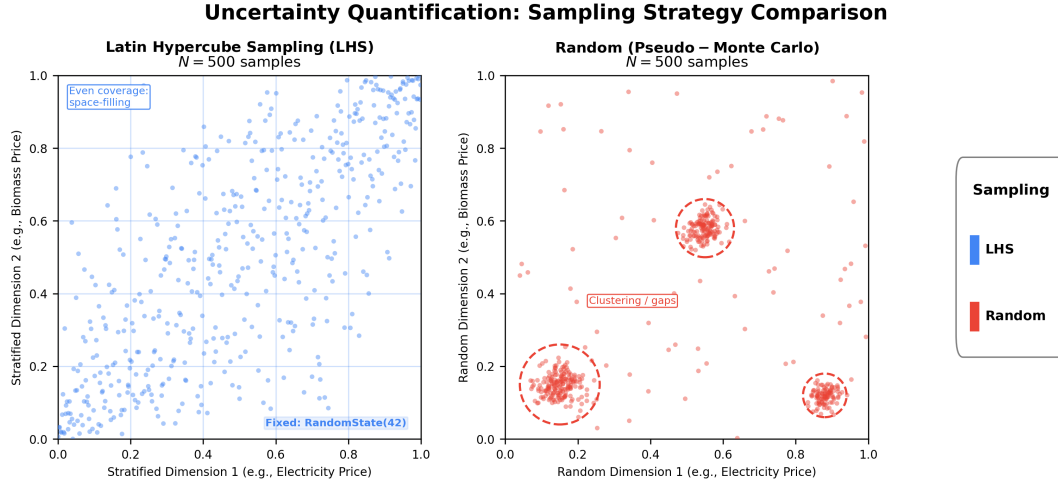


Figure 5: Comparison of sampling strategies for Monte Carlo uncertainty propagation with $N = 500$ realizations. Latin Hypercube Sampling provides more even coverage of the uncertainty space than conventional random sampling, reducing clustering and gaps in sampled parameter combinations.

Table 12

Deterministic decision trace for ST010.

Decision step	Evidence used	Outcome	Interpretation
Gate 1: DH feasibility	$\Delta p_{\text{spec}} = 53.0 \text{ Pa/m}$; $v_{\text{max}} = 0.95 \text{ m/s}$	PASS	DH remains admissible
Gate 1: HP feasibility	Maximum feeder loading 201.5%; 12 overloaded lines	FAIL	HP rollout requires grid reinforcement
Economic evaluation	DH LCOH = 64.64 EUR/MWh	DH evaluated	HP excluded because Gate 1 fails
Emissions evaluation	DH = 141.18 kgCO ₂ /MWh	DH evaluated	HP excluded because Gate 1 fails
Robustness evaluation	$\omega_{\text{DH}} = 1.00$	Stable	Monte Carlo sampling supports the DH recommendation
Final recommendation	Feasibility-first cascade over KPI contract	DH	District heating is the only admissible baseline pathway

5.6. Deterministic decision trace

Table 12 reconstructs the deterministic decision trace from the KPI contract. The first gate evaluates technical feasibility. DH passes the hydraulic feasibility gate, while HP fails the low-voltage grid-hosting gate because of feeder overloads. Since only one pathway remains technically admissible, the cascade resolves the recommendation before needing a direct feasible-versus-feasible cost comparison.

The final recommendation for ST010 is therefore district heating under the baseline assumptions. This result should not be interpreted as a universal rejection of heat pumps. Rather, it shows that for this street cluster and this low-voltage-grid representation, full heat-pump rollout is not technically admissible without reinforcement. A reinforced-grid HP scenario could be evaluated as a separate what-if case if reinforcement cost, routing, and implementation assumptions are added to the KPI contract.

Table 13

Internal ClaimExtractor validation results.

Claim type	Precision	Recall	Cases
LCOH numerical	1.00	0.86	14
CO ₂ comparative	0.94	0.94	12
Pressure / velocity	0.91	1.00	10
Win fraction	0.83	1.00	6
Overall	0.93	0.93	50

5.7. Explanation-safety validation

The constrained explanation layer was evaluated using internal safety-regression tests. The first stage, a deterministic ClaimExtractor, identifies quantitative claims in generated explanations and checks them against the KPI contract. The tested claim families include lifecycle cost, emissions, hydraulic indicators, Monte Carlo win-fraction statements, and binary feasibility claims.

The ClaimExtractor achieved approximately 0.93 precision and 0.93 recall on the internal synthetic test suite. These results show that deterministic claim checking is

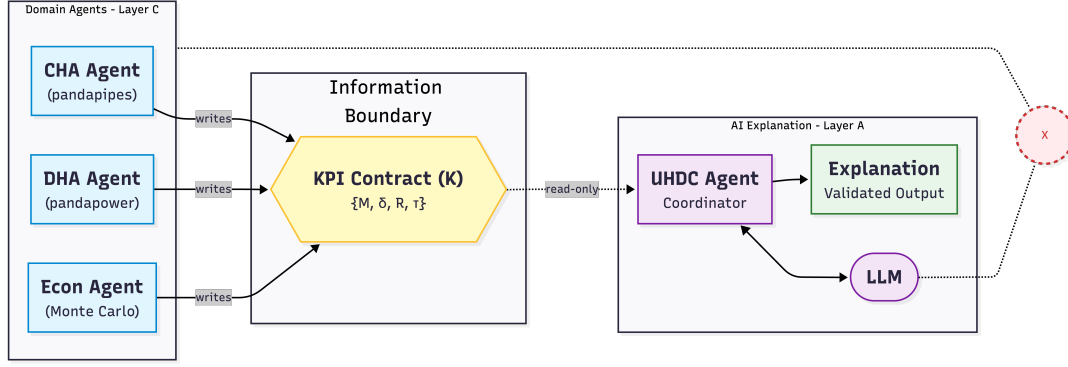
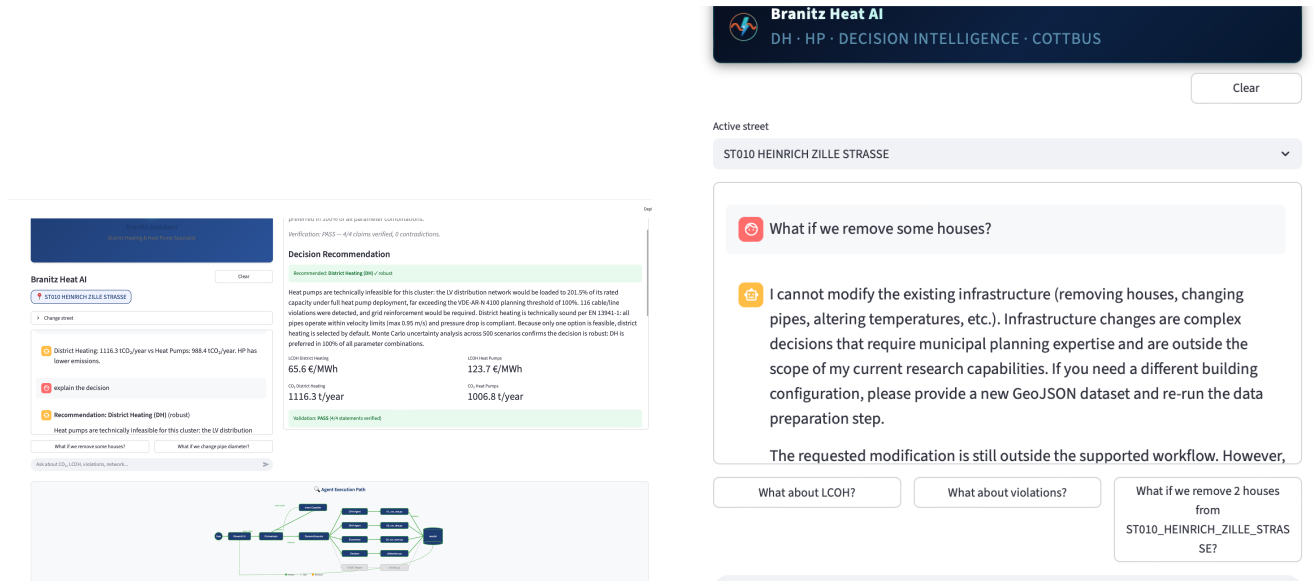


Figure 6: KPI-contract information boundary between deterministic domain agents and the constrained LLM-based explanation layer. Domain agents write validated metrics to the immutable KPI contract $\mathcal{K}$, while the Urban Heat Decision Coordinator has read-only access for explanation generation.



(a) Decision explanation for ST010 showing the district-heating recommendation and claim-verification status.

(b) Unsupported what-if query where the assistant refuses to modify infrastructure assumptions without a new dataset or scenario run.

Figure 7: Example constrained LLM interactions in the Branitz Heat AI prototype. (a) The assistant explains the ST010 district-heating recommendation using validated KPI fields and reports claim-verification status. (b) The assistant responds conservatively to an unsupported what-if request, avoiding unverified changes to buildings, pipes, or infrastructure assumptions. These examples illustrate the read-only explanation pattern used by the Urban Heat Decision Coordinator.

useful for catching unsupported numerical statements before they reach the user interface. However, the results should be interpreted as internal safety-regression evidence, not as an external benchmark. The remaining false positives and false negatives motivate the second-stage contradiction check using textual entailment, which is intended to block statements that contradict the KPI contract.

The key result for the journal paper is not that the LLM is intrinsically reliable. The key result is that the LLM is structurally constrained: the final recommendation is produced by deterministic rules over validated KPI fields, and the language model is used only to explain the resulting decision

trace. This design reduces the risk that unsupported natural-language claims can alter the engineering recommendation.

6. Discussion

The results demonstrate that the proposed framework can transform a multi-domain heating-planning problem into an auditable sequence of simulation, validation, decision, and explanation steps. The main contribution is not the selection of district heating for one street cluster alone, but the architecture through which that recommendation is produced. By coupling district-heating hydraulics, heat-pump low-voltage grid hosting, lifecycle cost, emissions, uncertainty analysis, and constrained explanation through a

shared KPI contract, the framework makes explicit why a pathway is recommended and which constraints determine the outcome.

6.1. Value beyond heat-density screening

The ST010 case illustrates why heat-density screening alone is insufficient for municipal heat planning. A conventional heat-density approach can indicate whether a cluster may be attractive for district-heating expansion, but it cannot determine whether a district-heating network satisfies hydraulic constraints or whether a heat-pump rollout can be hosted by the low-voltage grid. The proposed framework addresses this limitation by evaluating both pathways through explicit physical constraints before applying economic ranking.

This feasibility-first structure changes the interpretation of the result. The recommendation is not simply that district heating has a lower lifecycle cost under one parameter set. Rather, the evidence chain shows that the heat-pump pathway is technically blocked under the baseline low-voltage grid assumptions, while the district-heating pathway remains hydraulically admissible. This distinction is important for municipal decision-making because implementation feasibility can dominate nominal economic attractiveness. A pathway that appears cost-competitive in an unconstrained comparison is not a deployable planning option if it requires unmodelled grid reinforcement or violates engineering limits.

6.2. Interpretation of the ST010 recommendation

For ST010, the deterministic decision cascade selects district heating because it is the only pathway that passes the baseline feasibility gates. The Centralized Heating Analysis Agent reports district heating as hydraulically admissible, while the Decentralized Heating Analysis Agent identifies severe low-voltage feeder overloading under full heat-pump deployment. The final recommendation is therefore driven first by technical feasibility and only secondarily by lifecycle cost and emissions.

The ST010 result should therefore be interpreted as a feasibility-gated baseline assessment rather than as a complete investment comparison between district heating and grid-reinforced heat-pump deployment. Under the baseline assumptions, DH is the only technically admissible pathway for ST010. The DH pathway therefore proceeds to lifecycle-cost and emissions assessment, while the baseline full-HP rollout is excluded because it violates the adopted grid-hosting criteria.

This interpretation should be stated carefully. The result does not imply that heat pumps are generally unsuitable for Brantitz, Cottbus, or similar residential districts. It means that, under the modelled ST010 demand, grid representation, simultaneity assumptions, and heat-pump rollout scenario, the decentralized pathway cannot be recommended without explicit grid reinforcement. Heat pumps may become preferable in clusters with lower coincident heat demand, stronger low-voltage headroom, higher seasonal COP, lower

electricity prices, or where district-heating pipe length and connection cost are comparatively high.

The discussion therefore supports a conditional planning conclusion: district heating is robustly favored for ST010 under the baseline evidence chain, but the framework itself is technology-neutral. If reinforcement costs, partial heat-pump deployment, hybrid supply, building renovation, or alternative network routing are introduced as separate scenarios, the same decision logic can re-evaluate the recommendation.

6.3. Importance of deterministic orchestration

A central finding is that deterministic orchestration is essential when AI-supported workflows are applied to critical infrastructure planning. Municipal heat-planning recommendations must be reproducible, defensible, and explainable to technical and non-technical stakeholders. The framework addresses this requirement by separating domain agents and forcing all downstream decisions to operate on structured KPI fields.

The reproducibility results show that repeated execution produces stable KPI contracts under fixed inputs and fixed sampling configuration. This property matters because public-sector decisions may need to be reviewed months or years after an analysis was first produced. If the same scenario produces different recommendations across repeated executions, the tool would be difficult to defend in administrative or stakeholder processes. The use of deterministic preprocessing, schema-bound outputs, fixed Monte Carlo seeds, and content-addressable caching therefore contributes directly to auditability.

The cache-first implementation also has a practical benefit. Interactive planning workshops often involve follow-up questions, such as why a pathway was selected, which constraint failed, or how emissions compare. By reusing existing artifacts instead of re-running expensive simulations, the framework can support interactive exploration while preserving traceability. The thesis validation shows that multi-turn interaction and cache reuse substantially reduce response time while keeping the evidence chain inspectable.

6.4. AI safety through structural separation

The proposed framework treats the LLM as a constrained explanation component rather than as a simulator, optimizer, or decision-maker. This design choice is central to the contribution of the paper. In critical infrastructure planning, an unconstrained LLM could produce fluent but unsupported statements about pressure limits, grid violations, costs, or emissions. Such statements would be difficult for non-expert stakeholders to identify as incorrect.

The KPI-contract boundary reduces this risk. The LLM receives only validated fields from the deterministic workflow and is not allowed to invent numerical values, modify assumptions, or override the recommendation. The explanation layer therefore becomes a communication interface over computed evidence, not an autonomous reasoning authority. This is consistent with the broader requirement that AI tools

in energy and infrastructure contexts should remain reliable, auditable, and grounded in verifiable data.

The internal safety validation should also be interpreted conservatively. Claim extraction and contradiction checking provide useful regression tests, but they do not prove universal hallucination immunity. Their practical role is to reduce the probability that unsupported claims pass into the final explanation. The strongest safety property remains architectural: the recommendation itself is produced before the LLM is invoked, and the LLM cannot alter the underlying KPI contract or decision trace.

6.5. Practical implications for municipal planning

The framework has three practical implications for municipal heat planning.

First, it supports evidence-based pathway comparison. Instead of presenting district heating and heat pumps as generic alternatives, the framework identifies the binding constraint for each street cluster. In ST010, the binding constraint is low-voltage feeder capacity for the heat-pump pathway, not voltage range or transformer capacity. This enables planners to distinguish between pathways that are economically unattractive and pathways that are technically blocked.

Second, the KPI contract creates a common language between technical experts and decision-makers. Hydraulic indicators, grid-hosting indicators, LCOH, emissions, and robustness metrics are stored in one structured evidence object. This makes it easier to trace a recommendation back to specific inputs and constraints.

Third, the constrained explanation layer can make technical results accessible without allowing the language model to become the decision authority. This is useful for municipal contexts, where planners must often communicate complex infrastructure trade-offs to councils, utilities, and citizens.

6.6. Limitations and boundary conditions

The results should be interpreted within the boundaries of the current implementation.

First, the low-voltage grid representation is planning-grade rather than utility-calibrated. Exact distribution-system operator asset data were not available, so the heat-pump hosting analysis should be understood as a reproducible feasibility screen rather than a formal grid-connection study. Deployment-grade assessment would require validated feeder topology, cable types, transformer ratings, protection settings, and measured base-load profiles.

The case study does not include a reinforced-grid HP scenario. Consequently, the reported ST010 comparison should be interpreted as a baseline feasibility-gated assessment rather than a complete investment comparison between district heating and grid-reinforced heat-pump deployment.

Second, the district-heating layout is generated using reproducible street-following heuristics rather than global network optimization. This supports transparency and repeatability, but it may not identify the globally cost-minimal

or hydraulically optimal topology. Future work should compare heuristic layouts with optimization-based routing and pipe-sizing methods.

Third, the demand model and heat-pump assumptions are simplified. Building-level demand is represented through planning-grade annual demand and hourly profile logic, while heat-pump performance is represented through COP assumptions rather than detailed building-specific thermodynamic simulation. Renovation effects, occupant behavior, control strategies, and thermal storage are not fully represented.

Fourth, the case study focuses on one detailed representative street cluster. Although the preprocessing pipeline covers multiple Branitz clusters, the journal result section currently uses ST010 as the main evidence chain. A stronger future version should report aggregate results across all street clusters and identify where DH, HP, hybrid, or inconclusive outcomes occur.

Fifth, the explanation-safety evaluation is internal. The ClaimExtractor and contradiction checks demonstrate bounded safety behavior on the tested cases, but they should be extended with external benchmark datasets, expert review, and stakeholder-oriented evaluation of explanation usefulness.

6.7. Transferability and future work

The proposed framework is transferable because the architecture is not tied to a specific street, city, or heating technology. The required inputs—building demand, street geometry, infrastructure assumptions, cost parameters, and emissions factors—are common to many municipal heat-planning processes. The same structure could be applied to other German municipalities or adapted to other European contexts where district heating, heat pumps, and low-voltage grid constraints must be compared.

Future work should extend the framework in five directions. First, utility-grade low-voltage data should be integrated to validate the heat-pump hosting results against real network assets. Second, all Branitz street clusters should be evaluated and summarized in a district-level decision map. Third, hybrid strategies should be modelled explicitly, including partial heat-pump deployment, grid reinforcement, building renovation, and thermal storage. Fourth, district-heating topology generation should be compared against optimization-based alternatives. Fifth, the explanation layer should be evaluated with municipal planners and domain experts to measure whether constrained LLM explanations improve transparency, trust, and decision quality.

Overall, the ST010 case demonstrates a practically relevant proof of concept: deterministic, auditable orchestration can turn multi-domain urban heat analysis into a transparent planning workflow in which evidence, decision logic, and explanation remain separable and inspectable.

7. Conclusions

This paper presented a deterministic multi-agent AI framework for auditable urban heat-planning decisions. The

framework couples pandapipes-based district-heating simulation, pandapower-based heat-pump grid-hosting analysis, lifecycle cost and emissions evaluation, Monte Carlo robustness analysis, and constrained LLM-based explanation through a schema-validated KPI contract. The central design principle is that physics-based simulation and deterministic decision logic remain authoritative, while the LLM is restricted to read-only explanation of validated results.

The Branitz case study demonstrated the framework on the representative street cluster ST010, Heinrich-Zille-Straße. The district-heating pathway passed the planning-level hydraulic feasibility gate, with the specific pressure-drop and velocity indicators remaining within the adopted limits. In contrast, the decentralized heat-pump pathway failed the low-voltage grid-hosting gate because feeder loading exceeded the planning threshold, even though the voltage band remained admissible. The feasibility-first cascade therefore selected district heating as the baseline recommendation for ST010 under the modelled assumptions.

The techno-economic and robustness results further support the decision trace. District heating remained the only technically admissible pathway in the deterministic base case and achieved a stable recommendation across the Monte Carlo uncertainty analysis, with a district-heating win fraction of $\omega_{\text{DH}} = 1.00$ in the evaluated scenario set. These results show the importance of evaluating technical feasibility before lifecycle cost or emissions ranking: an option with potentially attractive building-level characteristics cannot be recommended unless the required network infrastructure can host it.

Methodologically, the paper contributes a transferable energy-AI architecture rather than only a local case-study result. The proposed KPI-contract structure makes the evidence chain inspectable, the deterministic decision cascade makes the recommendation reproducible, and the constrained explanation layer makes the output communicable to planners and stakeholders without allowing the LLM to become a hidden decision-maker. This separation of simulation, decision, and explanation directly addresses the trust and auditability challenges associated with applying AI to critical infrastructure planning.

The current implementation has limitations. The low-voltage grid representation is planning-grade rather than utility-calibrated; district-heating routing is generated through reproducible heuristics rather than global optimization, and the detailed numerical evidence in this paper focuses on one representative street cluster. Future work should extend the analysis to all Branitz street clusters, integrate utility-grade low-voltage grid data, evaluate explicit grid-reinforcement and hybrid-heating scenarios, compare heuristic and optimization-based district-heating layouts, and test the explanation layer with municipal planners and domain experts.

Overall, the results indicate that deterministic multi-agent AI can support municipal heat planning when it is designed around verified simulators, explicit feasibility

gates, reproducible KPI contracts, and constrained natural-language explanation. The approach provides a transparent pathway for comparing centralized and decentralized heating options at street-cluster scale while keeping costs, emissions, infrastructure constraints, and decision rationale traceable.

CRediT authorship contribution statement

Ishantha Hewaratne: Methodology, Software, Data curation, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review and editing. **Tanja M. Kneiske:** Conceptualization, Supervision, Methodology, Resources, Project administration, Writing – review and editing. **Peter Langendörfer:** Supervision, Methodology, Writing – review and editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author(s) used ChatGPT for language editing, structural planning, and manuscript organization support. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

Funding information will be confirmed before submission.

Data and code availability

The source code and selected reproducibility materials are available in the project repository:

<https://github.com/ijhewaratne/multiagent-urban-heat-planning>.

Data availability is subject to the licensing conditions of the underlying geospatial and heat-demand sources.

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